

Providing Supplemental Programs for First Generation and/or Low-Income CS Students in Trying Financial Times

Lauren Bricker
(Moderator)
Javon Hickmon
University of Washington
Seattle, WA, USA
bricker@cs.uw.edu
javonh@cs.uw.edu

Lonnie Bowe
Concord University
Athens, WV, USA
lbowe@concord.edu

Jamie Payton
New Jersey Institute of
Technology
Newark, NJ, USA
jamie.payton@njit.edu

Kyrie Dowling
University of Pennsylvania
Philadelphia, PA, USA
kyrdo@seas.upenn.edu

Abstract

As more colleges and universities develop programs to support first-generation or low-income undergraduate Computer Science and Engineering students, retention and completion rates have empirically risen for students within such groups [9]. Despite this success, financial support for these programs has become increasingly uncertain due to the recent shift in United States government priorities, tightening of corporate coffers, and state budget shortfalls. This panel will probe the experiences of four faculty from diverse institutions regarding the fundamental components that have been successful in their programs. We aim to clarify how institutions can provide quality support to first-generation and low-income students during these trying financial times.

ACM Reference Format:

Lauren Bricker (Moderator), Javon Hickmon, Lonnie Bowe, Jamie Payton, and Kyrie Dowling. 2026. Providing Supplemental Programs for First Generation and/or Low-Income CS Students in Trying Financial Times. In *Proceedings of the 57th ACM Technical Symposium on Computer Science Education V.2 (SIGCSE TS 2026)*, February 18–21, 2026, St. Louis, MO, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3770761.3777134>

1 Summary

Successful examples of programs designed to support first generation and low-income students in computer science and engineering (CSE) degrees include the University of Colorado Boulder Goldshirt [5], PATHS at CS@Mines [4], the STARS Computing Corps [8], and the six universities (including University of Washington (UW)) funded through the NSF Redshirt in Engineering Consortium [6]. The types of interventions provided to students to assist their transitions from high school to college and achieve higher retention/completion rates vary between these programs: some include only summer bridge courses [7], others also include dedicated academic advising, and still others provide supplemental workshops for their required core engineering, computer science, math, and sciences courses. Many are financially supported through the National Science Foundations Scholarships in Science, Technology, Engineering, and Mathematics Program (S-STEM) grants [2];

however, private donations, corporate funding, and state budgets are also common sources of support.

Even though these programs have proven effective, funding has become increasingly uncertain given the recent shift in priorities by the United States government, tightening of corporate coffers, and state budget shortfalls. This panel includes faculty from four different institutions who will describe core components of their program, then explore ways to continue providing quality holistic support in lower resourced environments by answering the following:

- What do you think is the most vital part of your program that supports the students? Why did you come to this conclusion/what data do you have to back this up?
- What do you wish you could do if you have more resources such as time, money, people, or physical space?
- What challenges have you and your program(s) faced both pedagogically and financially?
- What challenges do you and your program(s) face now in the current political and financial landscape?
- What do you see as the best, worst, or reasonable case for what happens to your program in the next two years?
- What would be the bare minimum you could provide to support your students?

This panel's intended audience include those currently running similar programs who want to brainstorm how to sustain high-quality support amid shrinking budgets and those seeking low-cost strategies to launch such programs at their own institutions.

2 Panel Structure

The first ten minutes of this panel will include an introduction describing the needs and goals of programs to support first generation and low-income CSE students and audience polling to ascertain how many attendees are first generation/low income, and whether their institution has, or is interested in developing, a first generation or low-income student support program. A poll will be left open to allow attendees to submit questions during the presentations.

Each panelist will have five minutes to describe their institution's support program for first generation and low-income CSE students. For the remaining forty minutes, the moderator will facilitate a question and answer session consisting of the aforementioned questions, alongside live and polled audience questions.



This work is licensed under a Creative Commons Attribution 4.0 International License. *SIGCSE TS 2026, St. Louis, MO, USA*

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2255-4/2026/02

<https://doi.org/10.1145/3770761.3777134>

Notes on the panelist and attendee responses, along with any ideas brainstormed within the session, will be taken and disseminated to interested parties after the session.

3 Lauren Bricker (Moderator)

Dr. Lauren Bricker is an Associate Teaching Professor in the University of Washington (UW) Paul G. Allen School of Computer Science & Engineering. She has been instrumental in developing bridge and workshop courses for the State Academic Redshirt Student (STARS) [3] Program and Allen School Scholars Program (ASSP) [1].

4 Javon Hickmon

The UW Allen School has championed the Allen School Scholars Program (ASSP) to support first-generation and low-income students through a four week bridge course, supplemental workshops, advising, and mentorship. ASSP students begin in a bridge course designed to build a cohort then transition to regular courses with extra workshops that focus on reinforcing concepts with one-on-one meetings between each student and a Teaching Assistant.

Javon Hickmon is a Master's student in Computer Science at UW, and the instructor for ASSP supplemental workshops. He began his undergraduate studies as a student within ASSP, then became a workshop TA and led multiple efforts through an undergraduate student group to broaden participation within computing. He is currently an instructor for the workshops.

5 Lonnie Bowe

Concord University's SEAM (STEM Enrichment for Appalachia through Mentoring) program supports high-achieving, low-income students majoring in Computer Science, Computer Information Systems, Chemistry, and Environmental Geosciences. Funded by the NSF S-STEM program, SEAM provides four-year scholarships and supplemental activities designed to improve retention, academic success, and professional identity. As a rural, public institution in southern West Virginia, Concord serves a student body that is primarily first-generation and low-income, many from Appalachian coalfield communities.

Lonnie Bowe is an Assistant Professor of Computer Science at Concord University and a Co-PI on the SEAM project. As a faculty member working in a rural, resource-constrained environment, he is especially interested in exploring sustainable, low-cost approaches to student support.

6 Jamie Payton

Dr. Jamie Payton leads the STARS Computing Corps NSF Broadening Participation in Computing Alliance. Operating as a community of practice, the STARS Computing Corps engages faculty and students with a shared commitment to broadening participation in computing. The STARS Computing Corps has engaged 70+ colleges and universities in projects designed to promote persistence in computing degree programs by connecting computing to community and societal impacts, reinforcing computing knowledge through teaching and practice, and developing professional, entrepreneurial, and creative skills. Student members have positive gains in GPA, self-efficacy, computing identity, and commitment to computing, with enhanced outcomes for women, African American/Black and

Hispanic/Latinx students. Payton has also led the INTuiTN NSF S-STEM Research Hub, a national project that seeks to investigate non-tuition needs among financially disadvantaged community college STEM students, and the NSF S-STEM Sci+Tech Scholars project, which aims to democratize access to tech-enabled science education for financially disadvantaged undergraduate students.

Jamie Payton is a Professor of Computer Science and Dean of the Ying Wu College of Computing at the New Jersey Institute of Technology. Her research interests include broadening participation in computing and evidence-based approaches to improving computer science education.

7 Kyrie Dowling

The University of Pennsylvania's Pre-First Year Program (PFP) is a four-week intensive academic program targeting first generation and income-eligible students as well as students from large urban and small rural high schools. The program prepares students for introductory courses in a chosen school (e.g., the School of Engineering and Applied Sciences), in addition to providing connections to faculty, academic counseling, and peer mentors. PFP is offered at no additional cost to students who participate.

Kyrie Dowling is a Lecturer in Computer and Information Science at the University of Pennsylvania. She has co-instructed the PFP Engineering Programming Course.

References

- [1] [n. d.]. *Allen School Scholars Program*. Retrieved June 24, 2025 from <https://www.cs.washington.edu/allen-scholars/>
- [2] [n. d.]. *NSF Scholarships in Science, Technology, Engineering, and Mathematics Program (S-STEM)*. Retrieved June 26, 2025 from <https://www.nsf.gov/funding/opportunities/s-stem-nsf-scholarships-science-technology-engineering-mathematics>
- [3] [n. d.]. *STARS scholars*. Retrieved June 24, 2025 from <https://www.engr.washington.edu/stars>
- [4] Tracy Camp, Christine Liebe, and Heather Thiry. 2021. CS@Mines Successful S-STEM Scholarship Ecosystem for Low-Income and Underrepresented Students. In *Proceedings of the 52nd ACM Technical Symposium on Computer Science Education (Virtual Event, USA) (SIGCSE '21)*. Association for Computing Machinery, New York, NY, USA, 830–836. doi:10.1145/3408877.3432524
- [5] Tanya Ennis, Jana Milford, Beth Myers, Jacquelyn Sullivan, Diane Sieber, Daniel Knight, and Ann Scarritt. 2010. Goldshirt Transitional Program: Creating Engineering Capacity And Expanding Diversity Through A Performance Enhancing Year. In *2010 Annual Conference & Exposition*. ASEE Conferences, Louisville, Kentucky. <https://peer.asee.org/15886>.
- [6] Beth A. Myers, Emily Knaphus-Soran, Donna C. Llewellyn, Ann Delaney, Sonya Cunningham, Pamela Cosman, Tanya D. Ennis, Katherine Christine Tetrick, Eve A. Riskin, Janet Callahan, and Kevin Pitts. 2018. Redshirt in Engineering: A Model for Improving Equity and Inclusion Paper. In *2018 CoNECD - The Collaborative Network for Engineering and Computing Diversity Conference* (Crystal City, VA, USA). American Society for Engineering Education, New York, NY, USA. doi:110.18260/1-2--29571 <https://peer.asee.org/29571>.
- [7] Abby O'Neill, Stella Kaval, Mallika Reddy, Alvaro Monge, Colleen Lewis, and Narges Norouzi. 2025. Launching and Enhancing Summer Bridge Programs. In *Proceedings of the 56th ACM Technical Symposium on Computer Science Education V. 2 (Pittsburgh, PA, USA) (SIGCSE 2025)*. Association for Computing Machinery, New York, NY, USA, 1725. doi:10.1145/3641555.3705094
- [8] Jamie Payton, Tiffany Barnes, Kim Buch, Audrey Rorrer, Huifang Zuo, Kinnis Gosha, Kristine Nagel, Nannette Napier, Ebrahim Randeree, and Lawrence Dennis. 2016. STARS Computing Corps: Enhancing Engagement of Underrepresented Students and Building Community in Computing. *Computing in Science & Engineering* 18, 3 (2016), 44–57. doi:10.1109/MCSE.2016.42
- [9] Eve A. Riskin, Jana Milford, Janet Callahan, Pamela Cosman, John B. Schneider, Kevin Pitts, Emily Knaphus-Soran, Donna C. Llewellyn, Ann E. Delaney, Myers Beth A., Katherine Christine Tetrick, Sonya Cunningham, Tanya D. Ennis, Kevin O'Connor, Michelle Ferrez, Tiffany D. Pan, and Jessica Baldis. 2018. Board 105: The Redshirt in Engineering Consortium: Progress and Early Insights. In *2018 ASEE Annual Conference & Exposition*. ASEE Conferences, Salt Lake City, UT. <https://peer.asee.org/29866>.